

# HapticDiveBuddy: Assessing utility of haptic feedback in navigating high turbidity diving environments

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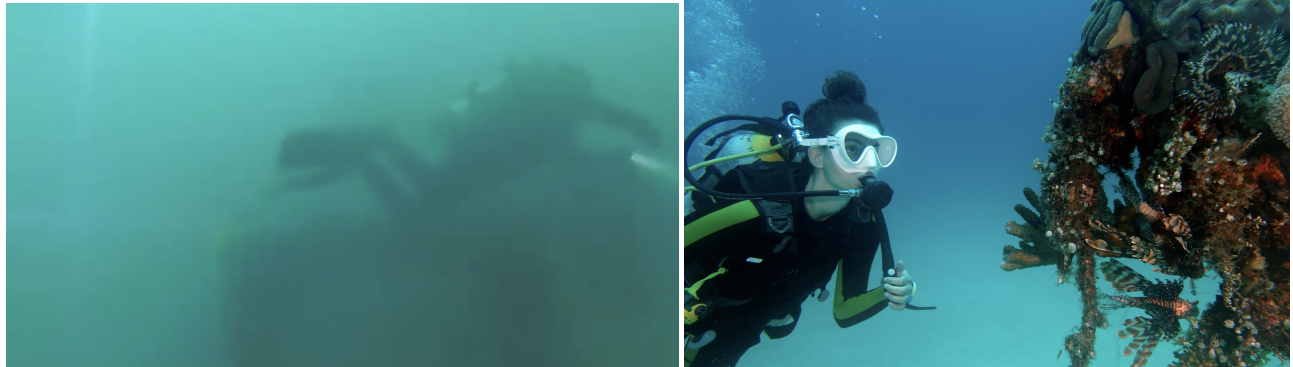


Figure 1: Same underwater obstruction at various levels of visibility (left) 2 meter visibility (right) 15 meter visibility

## ABSTRACT

Rescue, cave and police divers find themselves often working in murky, high turbidity environments. Those challenging environments often times are the scenes of rescue missions, key evidence recovery and in case of cave divers scientific discoveries. Work in those environments requires specialized training and despite rigorous safety protocols those environments become risky. Some of the main risks are disorientation, entanglement, injury as a result of collision with an unseen object and running out of gas as a result of prolonged stay underwater in attempt to orientate oneself. Therefore, the main focus of the proposed undergoing research is to prototype an assistive device for underwater navigation and test the suitability of haptic technology to underwater applications. The main contributions of the following project are (1) Proposal of a haptic diving costume technology navigation system for diving application (2) Conducting experiments aiming at assessing the potential strengths and drawbacks of haptic feedback in marine environments.

## CCS CONCEPTS

• Human-centered computing → Gestural input.

## KEYWORDS

Aquatic navigation, haptic feedback, marine signal processing

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## 1 MOTIVATION

As a trained diver with formal education in software and hardware technology, I believe that haptic technology has the potential to greatly enhance the diving experience and improve the safety and efficiency of diving operations. An early warning system could also contribute to the health of the underwater environment, as diver confrontation with an obstacle can be often times also detrimental to the marine environment. Furthermore, according to the literature review and interviews with professional divers haptic technology remains under-explored in marine research despite multiple examples of benefits of assistive technology in marine environments in riverbed mapping[6], robot-human navigation [4] and robot assisted archeological excavations [7].

## 2 RELATED WORK AND BACKGROUND

Current body of research on the subject of performance of haptic technology underwater is limited. Some of the most outstanding research was done by [5] and [2] from which it was concluded that haptic signal despite water signal absorption can successfully travel through water. The design of haptic actuators proposed in the aforementioned research however, differs from the research presented in this paper. Although previous authors have used pneumatic actuators and air bubbles to transmit signal the author of this paper makes the case that piezoelectric actuator technology can be used with the same degree of success in underwater applications.

Haptic actuators can be sub-divided into pneumatic and hydraulic actuators, piezoelectric, eccentric rotating mass and linear resonant actuators.

Piezoelectric haptics are typically made of a piezoelectric ceramic, which is used to generate vibrations or pressure changes in the water. When an electrical voltage is applied to the actuator, it expands or contracts, causing the surrounding water to vibrate or experience a pressure change. They are relatively durable and can withstand the harsh underwater environment. Furthermore, they can generate very high frequency vibrations, which can be useful for high-resolution imaging and sonar applications. [8] The limitation of piezoelectric haptics is their high power consumption and the need of a power source. This technology was already successfully used in underwater tactile applications. [1] [3]

Eccentric rotating mass (ERM) haptics use a small motor with an off-center weight (eccentric mass) to generate vibrations. The motor spins the eccentric mass, which causes the device to vibrate. ERM haptics are commonly used for touch screen interactions. ERM haptics are not suitable for underwater communication, as they require air to work and cannot operate in water. A Linear resonant actuator (LRA) uses a small motor to generate vibrations by rapidly moving a weight back and forth along a linear axis. The motor oscillates the weight at a specific resonant frequency, which causes the device to vibrate. However, LRAs can be sensitive to temperature changes. Like ERM haptics, LRAs require air to work, therefore the author predicts worse performance underwater.

### 3 PROBLEM STATEMENT AND RESEARCH CONTRIBUTION

Given the lack of research in haptic feedback underwater the proposed research aims at establish an understanding of good use practices of this technology in high turbidity environment navigation context. The problem that I am aiming to solve is safety of the marine life, as well as diver safety in emergency diving, which unlike recreational diving often times takes place in unsafe conditions. The research contributions are as follows:

- (1) Determine optimal placement of actuators and sensors
- (2) Evaluate haptic actuator feedback strength in 50l tank environment
- (3) Test the proposed system performance underwater at low depths

Therefore, the central research question that this research aims at answering is: "What is the utility and limitations of haptic feedback for divers in terms of improving navigation and enhancing safety in high turbidity marine environments".

### 4 METHOD

The proposed research can be sub divided into two research phases. First, the dry-environment and second, wet-environment studies. Although the author sees limitations of the dry-environment such as vertical orientation of the body in opposition to horizontal position in wet-environments the utility of testing the technology first outside wet-environments with regards to number of sensors, potential design miss judgements and user feedback cannot be understated. Similarly, the wet-environment experiments can be sub-divided into fish tank experiments and swimming pool experiments.

#### 4.1 Dry-environment experiments

To test the placement and the number of required haptic actuators a dry-environment study has been conducted. In this study 5 participants were blindfolded and asked to navigate an obstacle environment from point A to B while wearing (1) no detector-actuators system (2) two detector-actuators placed on the outside of both arms (3) four detector-actuators placed on the outside of legs and arms. The design of the circuit that was used in dry-environment experiments and will be also used for the wet-environment experiment can be seen of the figure 3. The distance detectors used were the ultrasonic waterproof JSN-SR04T type and the haptic actuator used for dry-experiments was a ERM NFP-C1020 model. They actuators and sensors were placed within distance of 4 cm from each other. Furthermore, the shape of the obstacle was supposed to imitate a narrow passage that divers often time have to pass via marine trenches or narrow limestone passages underwater. This particular shape of the course was also chosen, because this is one of the rare instances where more vertical diving position can be assumed, thus the dry-experiment can mimic closer the diver course. The obstacles on the course were created using cardboard barriers in a large open space (university parking lot). The longitude and latitude of the participants was tracked using a custom-built iOS application which used Kalman Filters for improved GPS course tracking accuracy.

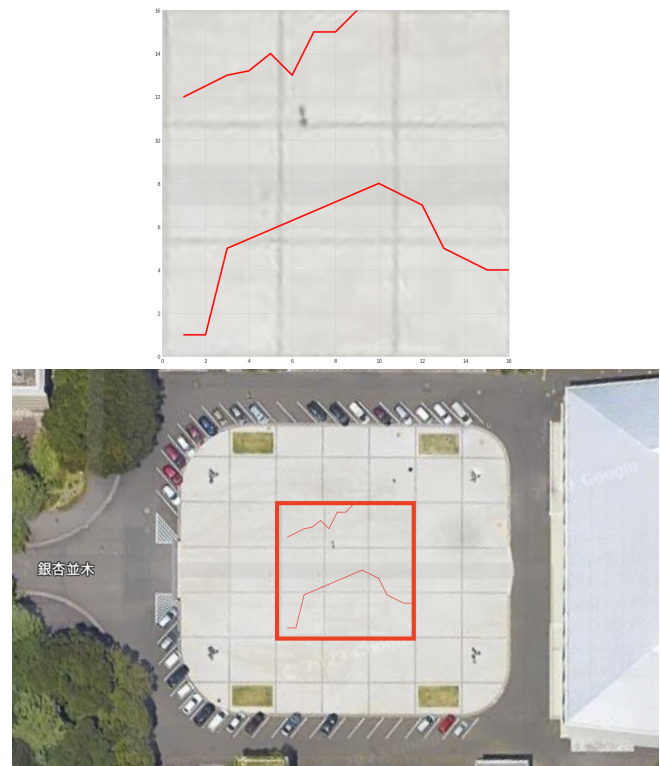
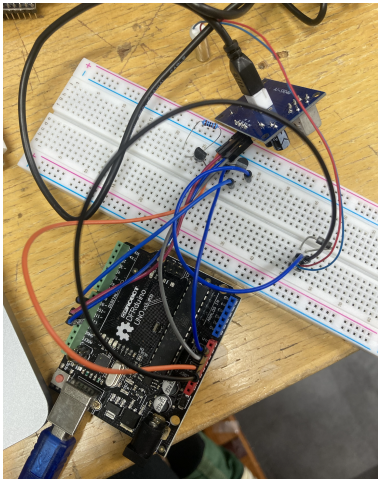
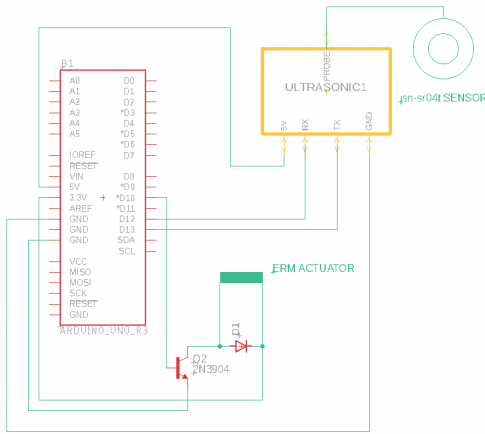


Figure 2: Dry-environment navigation task, where red lines mark obstacles

## 4.2 Wet-environment experiments

First, the proposed research evaluates the suitability of four popular haptic actuators with regards to their suitability in a wet-environment by placing them inside a 50l tank and asking 10 participants for subjective assessment of the strength of an actuator from the strongest (1) to the lowest(4). Given that this is a human-computer interaction research, prioritizing the diver experience, rather than choosing a quantitative study design, a subjective user study was selected. This is a still ongoing research and the swimming pool tests have not yet been executed. However, the next step for the wet-environment research will be to replicate a dry-environment navigation task in a shallow swimming pool using supervised snorkeling with a lifeguard for the safety of the participants, as they navigate through an unknown environment with obstacles with limited visibility. For limited visibility a diving mask with impaired visibility will be used. Secondly, various waterproofing techniques of electronic devices such as silicon sealing, conformal coating and encapsulating will be tested.

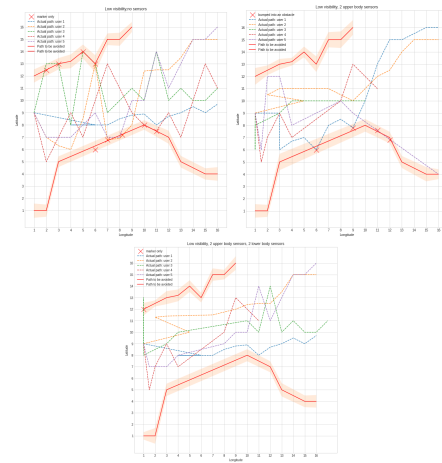


**Figure 3: Circuit design of distance sensor-haptic actuator system**

## 5 FINDINGS

### 5.1 Dry-environment findings

From the experiment the author concludes that including more sensors on the body is likely to have a positive impact on subject orientation at limited visibility condition. From the study the author also concludes that for some users the haptic feedback technology attached to their limbs was more intuitive than others and required shorter adaptation time to new technology. There was one disoriented user, despite having 4 sensors and 2 users which had to repeat the study due to disorientation with two sensors. The detailed paths that the participants took during the study are illustrated by Figure 4. Finally, from this study the author also learned that most of the participants when moving at limited visibility seem to have a reflex of extending their hands forward, which could also be relevant aspect to test regarding sensor placement, as well as, testing whether a similar behavior can be observed in wet-environments.



**Figure 4: Dry-environment navigation task data collection from the left (1) No sensors attached (2) 2 sensors attached to the upper body (3) 4 sensors attached, 2 to the upper body and 2 to the legs**

### 5.2 Wet-environment findings

For the initial wet-environment experiments various haptic actuators were submerged in a 50liter water tank and 10 participants were asked to order the haptic actuators from the strongest tactile sensation (1) to lowest (4). Table 1 summarized the subjective Haptic perception test outcomes for the 10 users. Given that most of the participants ranked piezo sensor as the strongest one under same voltage conditions this is the haptic sensor that will be used in further swimming pool test.

### 5.3 Conclusion

The ongoing study in haptic interaction underwater aims to contribute to the field of Human-Computer interaction by studying a haptic technology in novel environment and application. The author hopes that the experiments proposed such as dry-environment

| Haptic perception test |                                    |                             |                                       |                                            |
|------------------------|------------------------------------|-----------------------------|---------------------------------------|--------------------------------------------|
| Participant            | tdk piezo powerhap<br>0909H011V060 | ERM coin motor<br>NFP-C1020 | Encapsulated DC<br>Motor<br>NFP-E1015 | Linear Vibration<br>Motor<br>NFP-ELV061228 |
| Person 1               | 1                                  | 2                           | 3                                     | 4                                          |
| Person 2               | 1                                  | 2                           | 4                                     | 3                                          |
| Person 3               | 3                                  | 2                           | 1                                     | 3                                          |
| Person 4               | 2                                  | 3                           | 2                                     | 4                                          |
| Person 5               | 1                                  | 2                           | 4                                     | 3                                          |
| Person 6               | 2                                  | 1                           | 4                                     | 3                                          |
| Person 7               | 1                                  | 2                           | 4                                     | 3                                          |
| Person 8               | 1                                  | 2                           | 4                                     | 3                                          |
| Person 9               | 2                                  | 1                           | 4                                     | 3                                          |
| Person 10              | 1                                  | 2                           | 4                                     | 3                                          |

Table 1: haptic perception test

restricted visibility navigation task will contribute to establishing understanding of best practices for delivering haptic feedback. Furthermore, the wet-environment tank experiment provides the CHI community with a user-generated understanding of haptic actuators feedback. The author of this study hopes that the future swimming pool experiment replicating the navigation task from the dry-environment experiment proposed, alongside interviews with professional divers would give a concluding answer to the posed research question. Finally, the proposed research aims at creating a positive impact on human diving safety and marine environment conservation by establishing less invasive diving practices in low visibility environments.

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